

BISMA ABBASI

SOFTWARE ENGINEER • BACKEND DEVELOPER • AI INTEGRATION SPECIALIST

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PROFESSIONAL SUMMARY

Results-driven **Software Engineer** specializing in **Backend Architecture**, **RESTful API Development**, and **AI/LLM Integration**. Proficient in **NestJS**, **TypeScript**, **Node.js**, and **MongoDB Atlas**, with production experience deploying scalable cloud systems featuring **JWT Authentication**, **Role-Based Access Control (RBAC)**, **Swagger/OpenAPI** documentation, and **CI/CD** pipelines. Proven track record integrating large language models (**Groq LLaMA 3.3**, **OpenAI API**, **Gemini API**) via advanced **Prompt Engineering** and **LangChain** into **Railway** and **Vercel** environments. Strong foundation in **Microservices**, **System Design**, and **Agile/Scrum** with a cumulative CGPA of **3.70 / 4.00**.

CORE TECHNICAL SKILLS

Languages	TypeScript, JavaScript (ES6+), Python, SQL, C++, Java
Backend Frameworks	NestJS, Node.js, Express.js, RESTful API Design, JWT Auth, RBAC Guards, Middleware, DTOs, Validation Pipes, Microservices
AI & LLM Integration	Groq LLaMA 3.3, OpenAI API, Gemini API, Prompt Engineering, LangChain, Generative AI, RAG, Data Analytics
Databases & Storage	MongoDB Atlas, Mongoose ODM, PostgreSQL, MySQL, Supabase, Firebase
Cloud & DevOps	Railway, Vercel, Azure Fundamentals, Docker, Git, GitHub, CI/CD, Postman, Swagger/OpenAPI, Multer
Core Concepts	OOP, Data Structures & Algorithms (DSA), System Design, API Security, Software Architecture, Agile/Scrum

TECHNICAL PROJECTS

AI Study & Interview Prep API	2025
NestJS, TypeScript, MongoDB Atlas, Groq LLaMA 3.3, JWT Authentication, Swagger/OpenAPI	
- Reduced preparation time by 70% by architecting a production-ready NestJS REST API that generates mock interview questions and structured model answers using Groq LLaMA 3.3 as validated JSON responses. - Achieved sub-2-second LLM response latency by engineering rigid, structured Prompt Engineering templates with strict output validation. - Secured 100% of endpoints by implementing robust JWT Authentication and Role-Based Access Control (RBAC) guards; documented via Swagger/OpenAPI.	
Users Management REST API	2025
NestJS, TypeScript, MongoDB Atlas, JWT, bcrypt, Railway, CI/CD Deployment	
- Deployed production REST API live on Railway with full CRUD operations, Mongoose schema validation, and startup automation scripts. - Eliminated password security vulnerabilities by implementing bcrypt hashing with stateless JWT authentication and token refresh flows. - Reduced request validation errors by 95% via DTO validation with class-validator and a global ValidationPipe middleware layer.	
ViewTube Backend API	2024
Node.js, Express.js, MongoDB Atlas, Cloudinary CDN, JWT Auth, Multer	
- Engineered modular video-sharing backend architecture processing media assets using MongoDB Aggregation Pipelines for complex data queries. - Reduced media upload latency by 45% by integrating Cloudinary CDN and Multer file-parsing stream pipelines for optimised asset delivery.	
AI-Driven Smart Pediatric Health Advisor (Capstone Project)	2024 – 2025
Python, Flask, AI Integration, LLM APIs, Agile/Scrum Methodologies	
- Appointed Project Team Lead (managed 3 developers) to deliver an AI-powered diagnostic healthcare advisor system on schedule using LLM APIs. - Directed 4 full Agile/Scrum sprints, coordinating task allocation to connect a Python service layer with a client application, achieving all sprint goals.	

Dynamic CSV Importer & Data Mapping System

2023

Node.js, Express.js, MongoDB, Muler, Data Stream Processing

- Accelerated batch data ingestion by 10× (60% time reduction) via concurrent MongoDB insertMany() pipeline execution with stream processing.
- Eradicated runtime schema-mismatch crashes by designing an algorithmic dynamic data mapping and validation layer for heterogeneous CSV schemas.

PROFESSIONAL EXPERIENCE

Pakistan Television Corporation (PTV)

Jul 2025 – Aug 2025

Software Engineer Intern • IT & Database Infrastructure

- Reduced data inconsistency by 40% by re-architecting relational enterprise database schemas and writing custom integrity-check procedures.
- Improved operational documentation coverage by 60% by authoring network workflow guides and standard operating procedures (SOPs).
- Maintained 99.9% system uptime through rigorous network monitoring, proactive troubleshooting, and infrastructure health checks.

Microsoft

Sep 2023 – Aug 2024

Microsoft Learn Student Ambassador • Technical Leadership (Remote)

- Upskilled 100+ students across 5 technical workshops on Python, Microsoft Azure Cloud fundamentals, and backend development.
- Accelerated onboarding speeds by 35% by developing clear technical learning guides for backend architecture and cloud deployment concepts.

EDUCATION

Bachelor of Engineering (B.E.) in Software Engineering

2021 – 2025

Quaid-e-Awam University of Engineering, Science & Technology • Pakistan

- Academic Performance: Achieved a top-tier cumulative CGPA of 3.70 / 4.00.
- Core Modules: Database Management Systems, Software Architecture, Object-Oriented Design, Data Structures & Algorithms.

PROFESSIONAL CERTIFICATIONS

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| • Meta Back-End Developer Professional Certificate • Meta / Coursera | • Google Advanced Data Analytics Certificate • Google / Coursera |
| • ChatGPT Prompt Engineering for Developers • DeepLearning.AI | • Generative AI for Everyone • DeepLearning.AI |
| • Back-End Development and APIs • freeCodeCamp | • MongoDB Basics — M001 Database Credential • MongoDB University |